

FCC-ee tau photon/pion reconstruction

Consolidated W-P8O diagnostic report

Generator/sample composition → tau topology → photon origin
→ PFO association → conditional PID → migration matrices

Date: 2026-09-01
Freeze status: **SCIENTIFIC ANALYSIS FROZEN AT 2026-09-01**
Scientific status: **PASS_WITH_WARNINGS**
Scope: documentation and provenance freeze only

The warnings are explicit event-record limitations, not failed integrity checks or pending software bugs.

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1 Executive summary

The large global W/P8O photon difference is driven primarily by sample/generator truth composition and truth-to-PFO association/reconstructability, not by a large change in conditional reconstructed photon PID. Tau topology, selected visible truth from tau decays, tau-decay photon kinematics, reconstructed PFO composition, and conditional PID are closely compatible between W and P8O.

The validated factorization is

origin/composition $\rightarrow$ phase space $\rightarrow$ unique PFO association $\rightarrow$ conditional PID $\rightarrow$ global outcome.

P8O contains 49,120 photons with positive stored genealogy evidence for `explicit_ISR`; they are exceptionally soft and beam-collinear. W contains 25,879 parentless non-tau photons without stored photon-to-beam ancestry. They remain `parentless_unresolved`; only “partially similar” or “ISR-like kinematics” is allowed. Direct tau daughters remain `tau_direct_daughter_unresolved`: the records do not support a positive-evidence, cross-generator FSR definition.

2 Samples, provenance, and frozen definitions

Table 1: Sample and event provenance.

Label	Generated sample	Reconstruction	Events
W	WHIZARD	our reconstruction	1,992,000 available; 18,000 detailed
P8C	PYTHIA8	collaborator reconstruction	migration baseline
P8O	PYTHIA8	our reconstruction	18,000 available and detailed

W versus P8O compares different generated samples under the same reconstruction chain. Detailed audits use 18,000 deterministic events per sample. Event-manifest SHA-256:

b2f81449579d33ec228b65c20a7927f8930ef19368aedd97244085c6e587cf3b.

P8O event `out_sim_edm4hep_1:838` has four terminal tau representatives and is retained as the sole inherited topology warning. All other 17,999 P8O events and all 18,000 W events have two.

Selected truth is frozen as implemented: `generatorStatus == 1`, neutrinos excluded, non-zero truth momentum, and the established species mapping. Reconstructed PID is $|\text{PDG}| = 11, 13, 22, 211, 310, 2112, 3122$ for electron, muon, photon, charged pion, K_S^0 , neutron, and Λ .

- **G**: geometric selected-truth association.
- **L_direct**: immediate detector-level MC contributor with frozen representative-PFO inversion.
- **L_ancestor**: nearest unique selected generator-level ancestor with the same inversion.

Tau origin is true exactly when recursively stored ancestry contains $|\text{PDG}| = 15$. G/L_direct/L_ancestor differences are *association-definition dependence*, not detector-efficiency variation.

Frozen `photon_origin_v1` contains the three tau categories, `explicit_ISR`, `parentless_unresolved`, the unresolved electron/positron/photon-chain categories, and `resolved_non_tau_other`. The full category names and precedence are frozen in the authoritative discovery report.

3 Global migration baseline

A truth-row-normalized migration matrix combines truth composition, PFO association/reconstruction, and conditional PID. Unmatched truth objects are not PID failures.

Table 2: Global photon migration headline.

Association	Sample	photon→photon	photon→unmatched
G	W	51.249 %	46.610 %
G	P8C	37.053 %	61.109 %
G	P8O	36.846 %	61.379 %
L_ancestor	W	51.149 %	45.068 %
L_ancestor	P8C	39.526 %	57.496 %
L_ancestor	P8O	39.355 %	57.786 %

The dominant difference is diagonal $\leftrightarrow$ unmatched, not migration to a specific wrong PID. Charged-pion outcomes are much more stable.

4 Tau topology and selected visible truth from tau decays

Table 3: Principal tau decay categories.

Category	W	P8O
1-prong with π^0	36.797 %	36.892 %
electron	18.336 %	17.688 %
muon	17.583 %	17.493 %
1-prong no π^0	12.256 %	12.144 %
3-prong no π^0	9.744 %	9.991 %
3-prong with π^0	5.153 %	5.664 %

Representative tau means are $p = 44.960/45.007$ GeV, $p_T = 33.142/33.279$ GeV, and $\theta = 89.961/89.983^\circ$ for W/P8O. The stored-tau-copy multiplicity plot is intentionally excluded from the scientific figures.

Both tau decay chains are included and neutrinos excluded. Selected visible tau-decay truth occurs at 5.350/event W and 5.334/event P8O. Photon, charged-pion, electron, and muon fractions are respectively 50.329/49.478 %, 34.701/35.444 %, 7.237/7.242 %, and 6.593/6.575 %. Tau-origin photons are 65.190 % of selected photons in the W 18k subset (65.481 % full-sample reference) and 49.163 % in P8O.

5 The key non-tau photon difference

Table 4: Non-tau photon population and kinematics.

Quantity	W	P8O
count	25,879	49,121
photons/event	1.438	2.729
mean E [GeV]	0.217	0.100
median E [GeV]	3.17×10^{-4}	1.41×10^{-6}
median p_T [GeV]	2.86×10^{-5}	9.88×10^{-9}
$ \cos \theta > 0.98$	50.191 %	83.748 %
$ \cos \theta > 0.999$	31.62 %	78.57 %

P8O has 1.898 times the non-tau photon multiplicity. Physical E and p_T are shown on logarithmic axes with logarithmically spaced bins, common W/P8O ranges, and explicit “/ log bin” normalization. No $\log_{10}(X)$ variable is used.

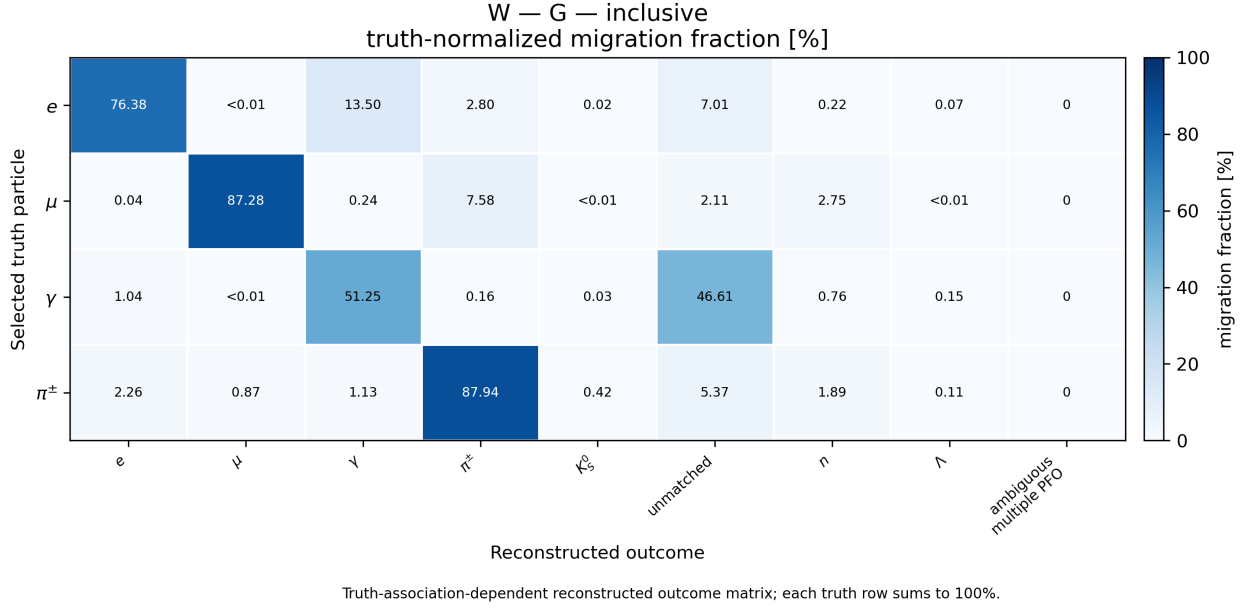


Figure 1: W inclusive G migration matrix. Rows are normalized independently to selected truth.

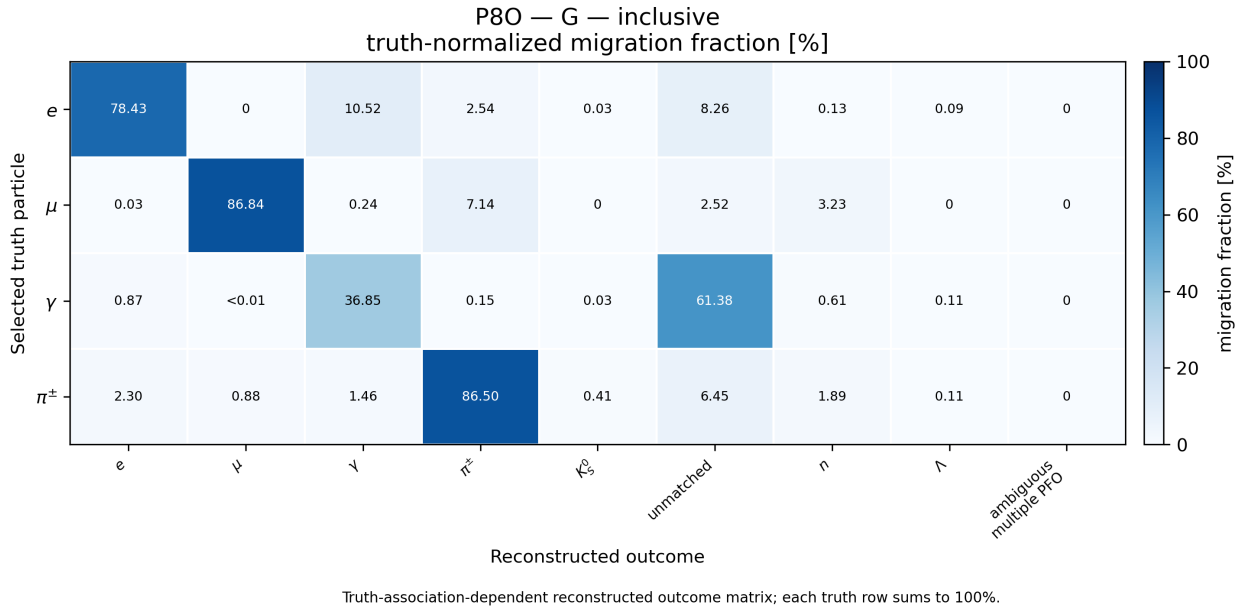


Figure 2: P8O inclusive G migration matrix, with the same definitions and scale as Fig. 1.

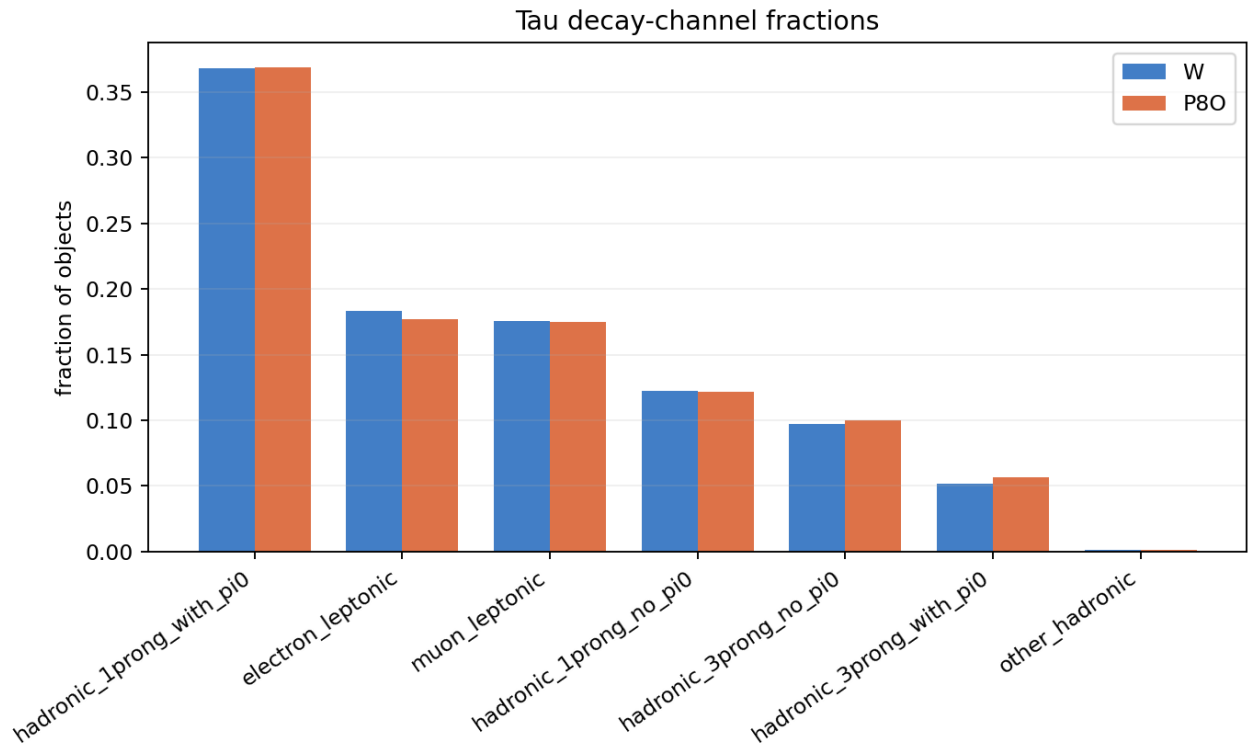


Figure 3: Tau decay channels. W and P80 show closely matching compact decay fractions.

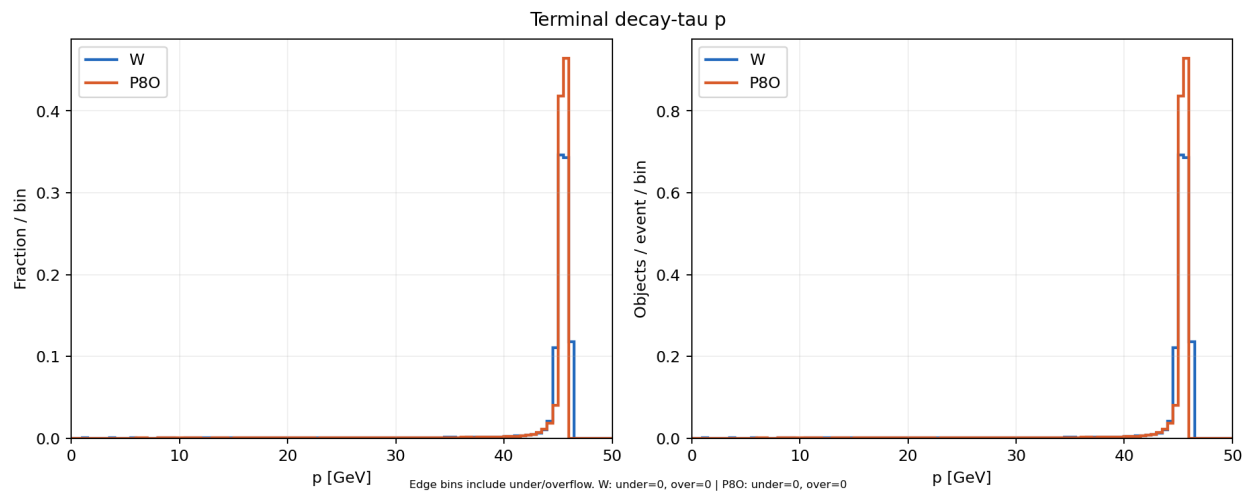


Figure 4: Representative tau momentum distributions.

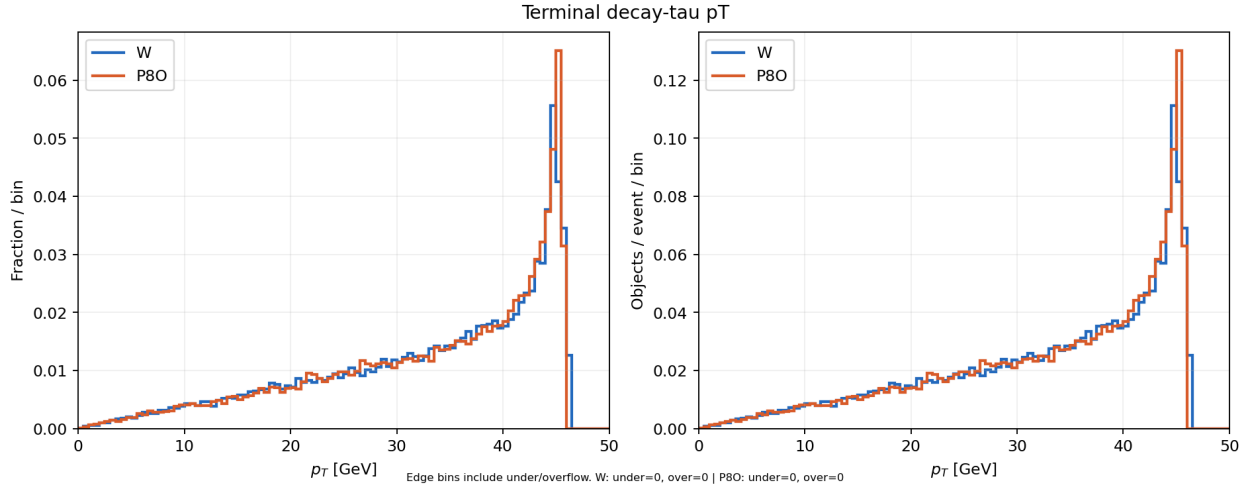


Figure 5: Representative tau transverse-momentum distributions.

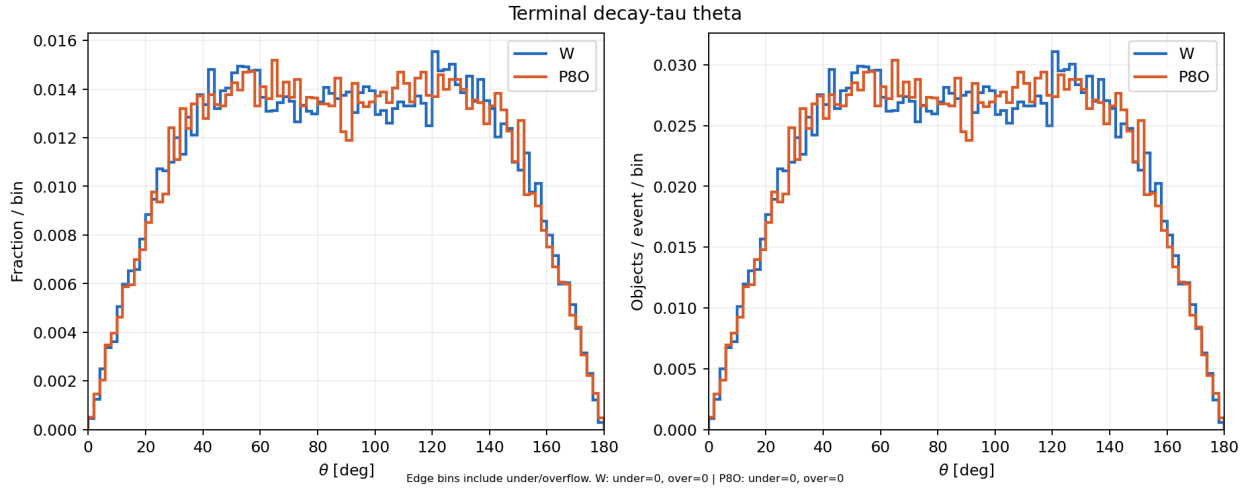


Figure 6: Representative tau polar-angle distributions.

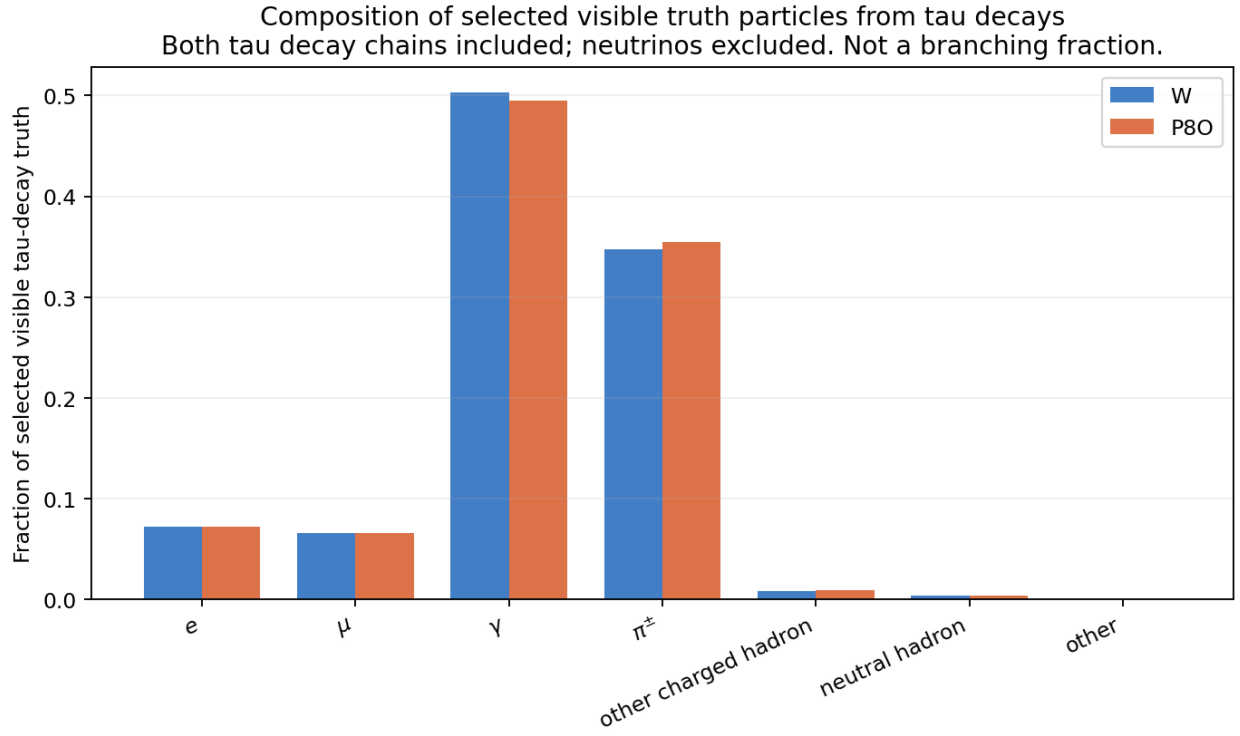


Figure 7: Composition of selected visible truth particles from both tau decays; neutrinos are excluded.

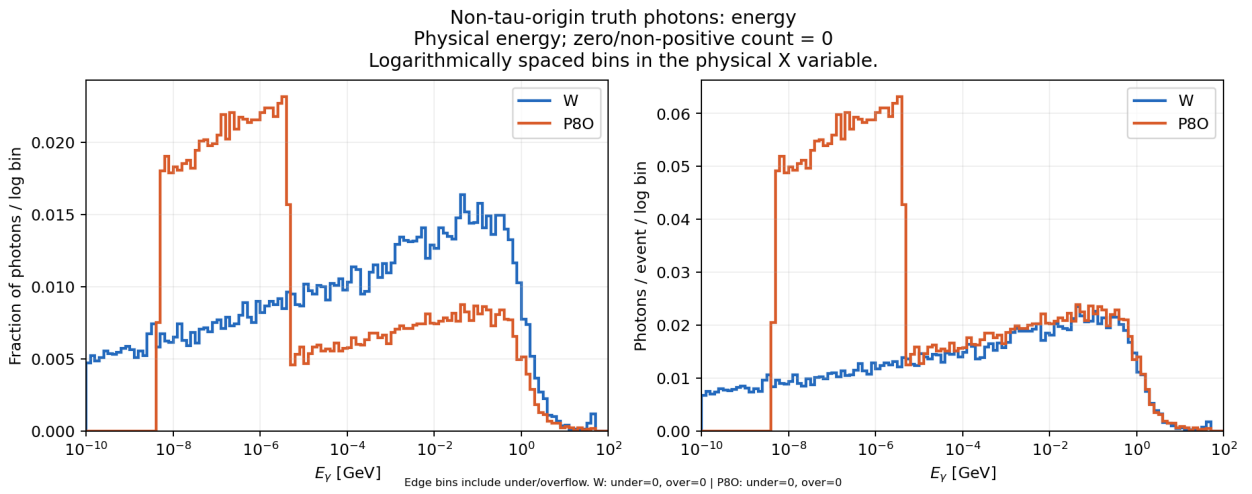


Figure 8: Non-tau photon physical energy. Left: independently unit-normalized shape. Right: yield per event. Logarithmically spaced bins in the physical variable.

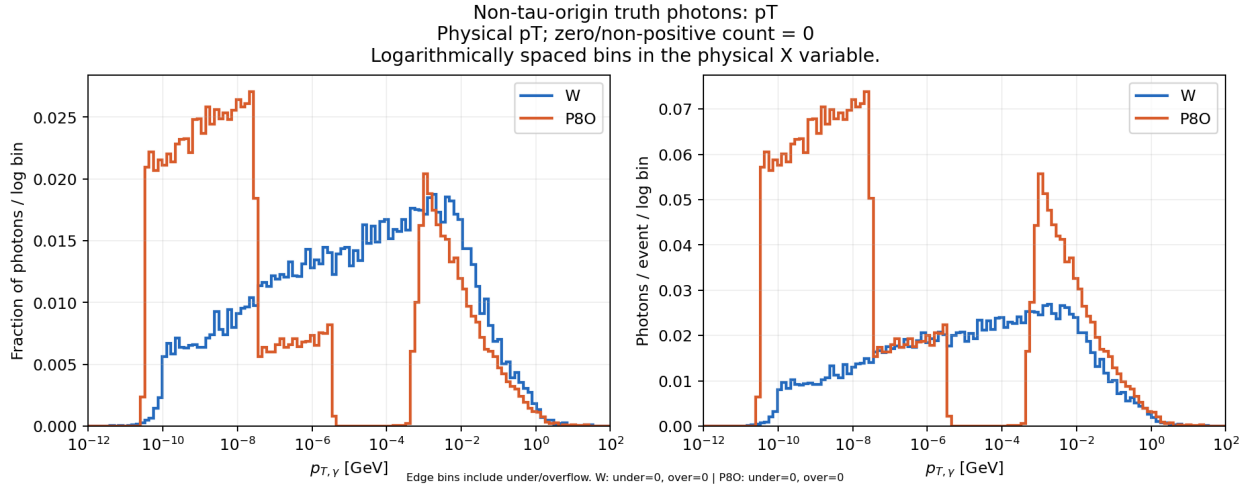


Figure 9: Non-tau photon physical p_T . Left: shape; right: yield. Logarithmically spaced bins in the physical variable.

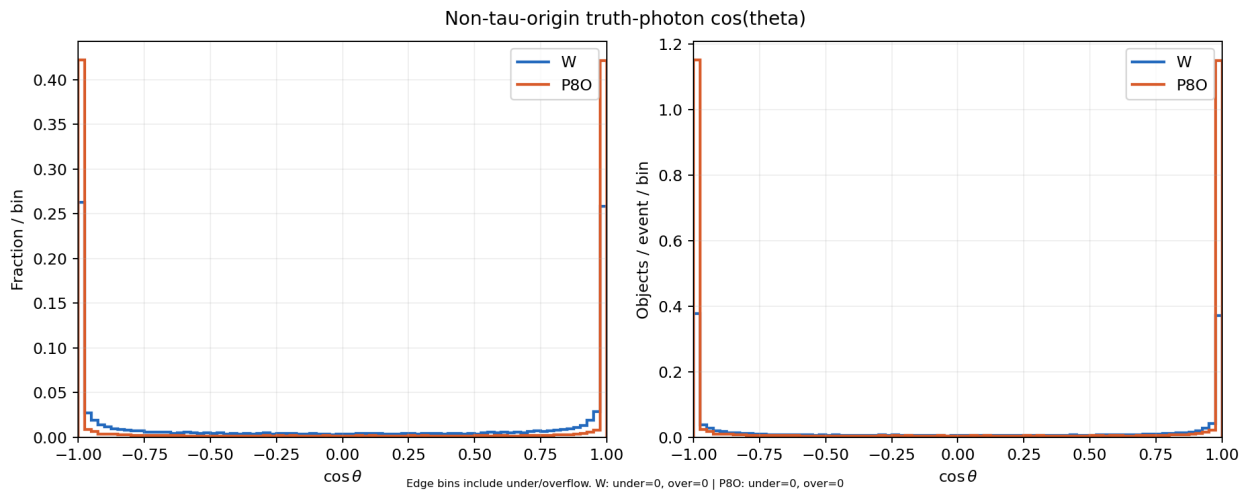


Figure 10: Non-tau photon angular topology. P80 is substantially more beam-collimated.

6 Reconstructed PFO content

Table 5: Reconstructed PFO PID yield per event.

PID	W	P8O	PID	W	P8O
electron	0.5403	0.5333	charged pion	1.8621	1.8847
muon	0.3265	0.3228	K_S^0	0.0152	0.0148
photon	2.8321	2.8820	neutron	0.4409	0.4573
			Λ	0.0182	0.0193

Despite 1.898 times more non-tau truth photons in P8O, photon-PFO yield differs by only 1.76 %. Charged/neutral populations and the principal reconstructed PID populations are closely matched.

7 Conditional PID

Conditional PID is $P(\text{correct reconstructed PID} \mid \text{unique associated PFO})$. Unmatched and ambiguous particles are excluded and are not PID failures.

Table 6: L_ancestor conditional PID accuracy.

Truth	W associated	W accuracy	P8O associated	P8O accuracy
electron	6,725	89.844 %	6,662	89.433 %
muon	6,254	88.647 %	6,185	88.747 %
photon	40,478	93.438 %	40,627	93.600 %
charged pion	32,447	92.267 %	32,908	91.965 %

All four accuracies agree within 0.42 percentage points. Mis-ID populations are often lower- p_T and, especially for pions, more forward/backward.

8 PFO association/reconstruction efficiency

Primary $\epsilon_{\text{unique}} = N(\text{unique associated PFO})/N(\text{selected truth})$ uses L_ancestor.

Table 7: L_ancestor truth-to-PFO association efficiency.

Species	W selected	W ϵ_{unique}	P8O selected	P8O ϵ_{unique}
electron	6,969	96.499 %	6,959	95.732 %
muon	6,349	98.504 %	6,313	97.972 %
photon	74,344	54.447 %	96,625	42.046 %
charged pion	33,416	97.100 %	34,030	96.703 %

Electrons, muons, and charged pions usually obtain a unique PFO. Photons are exceptional.

8.1 Tau-origin versus non-tau photons

Tau-origin efficiency is 81.013/81.873 % for W/P8O; non-tau efficiency is 4.695/3.530 %. For non-tau photons, W/P8O efficiencies are 0.313/0.230 % at $E < 0.1$ GeV, 1.632/1.243 % at $p_T < 0.1$ GeV, and 2.048/0.887 % for $|\cos\theta| > 0.98$. The P8O excess therefore occupies extremely low-efficiency phase space.

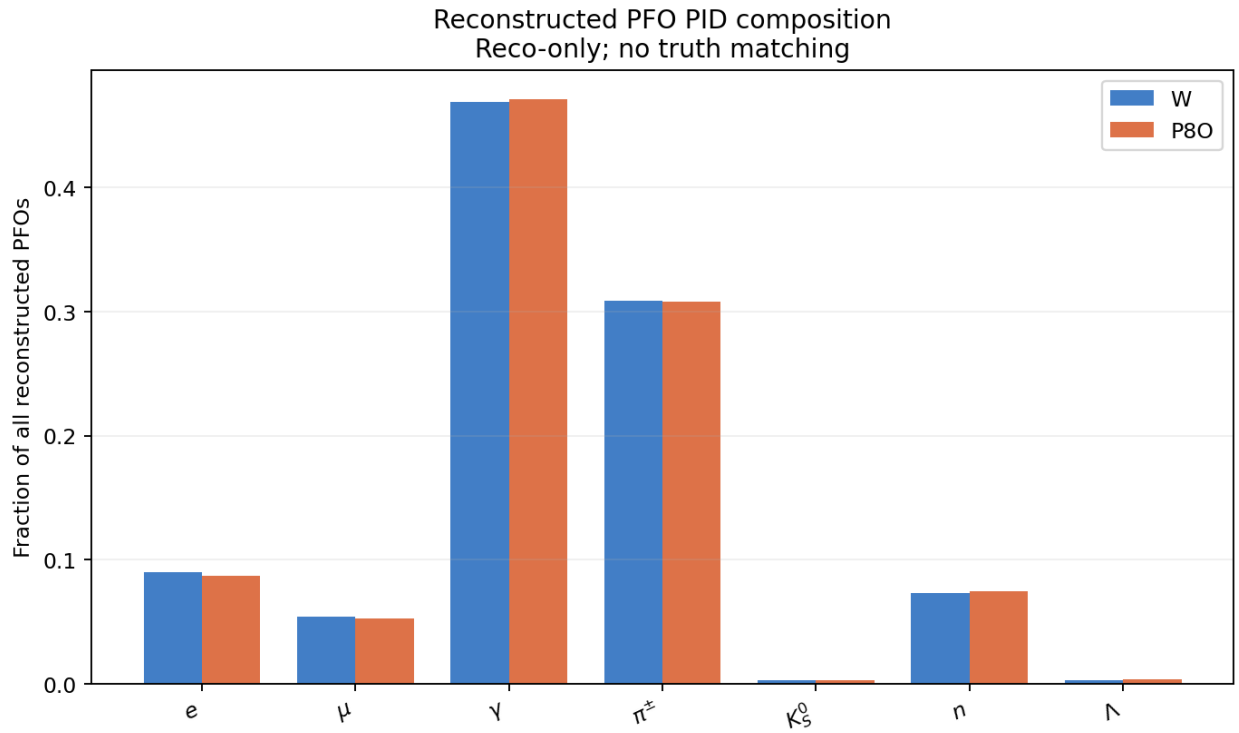


Figure 11: Reconstructed PFO PID composition.

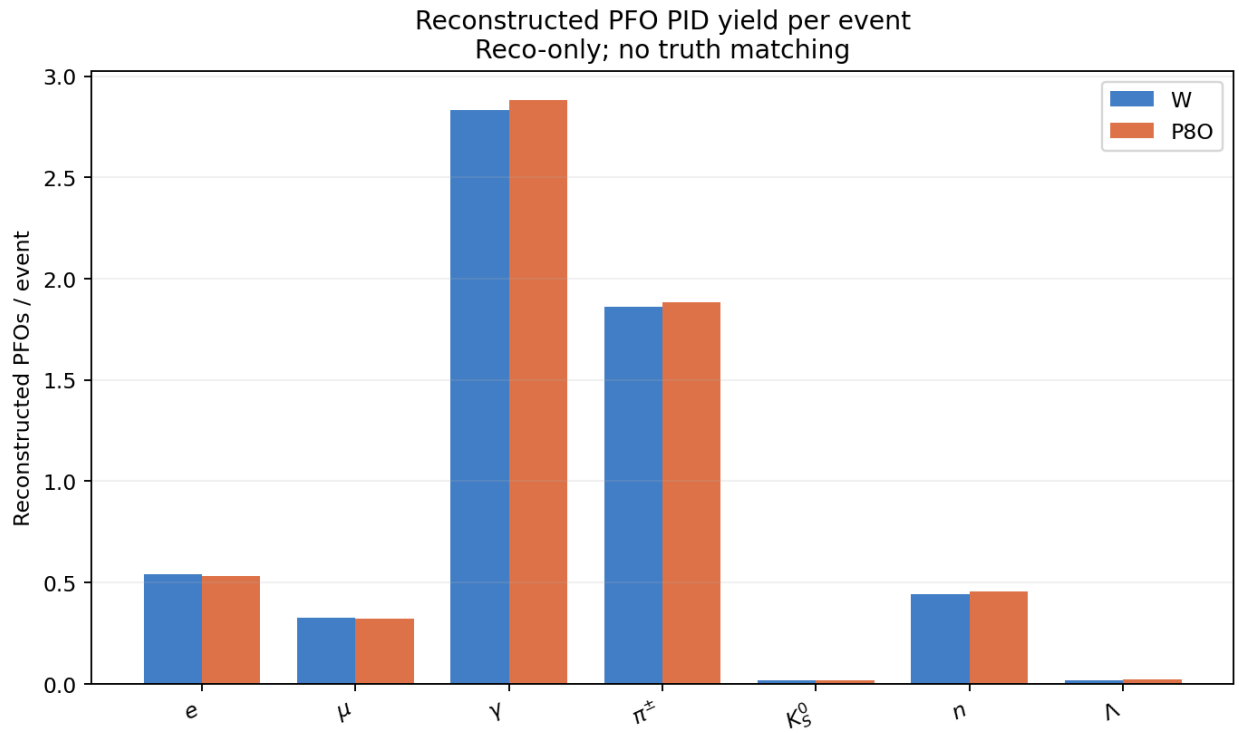
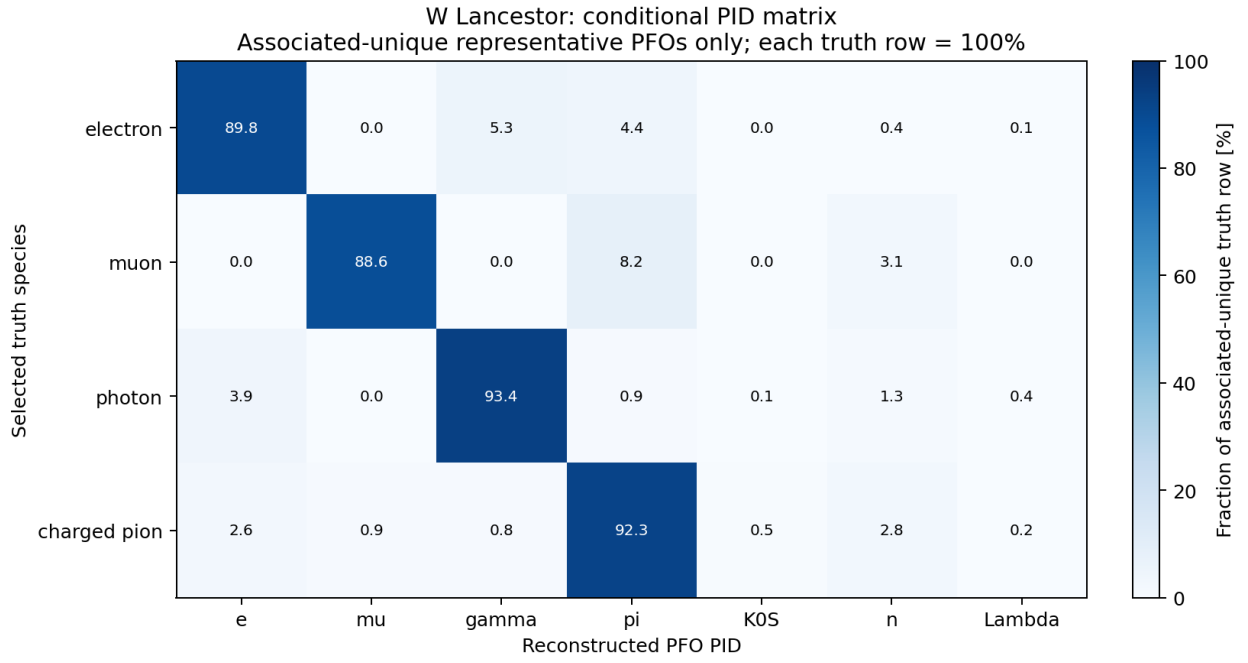
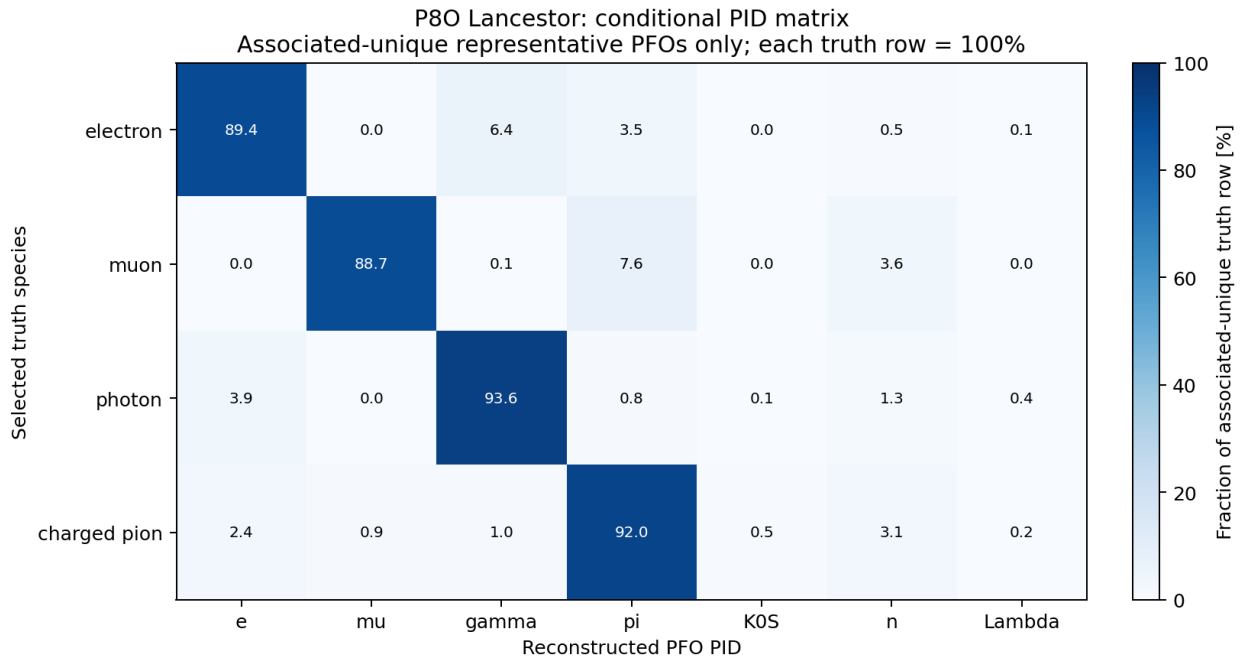


Figure 12: Reconstructed PFO PID yield per event.

Figure 13: W L_{ancestor} conditional PID matrix.Figure 14: P80 L_{ancestor} conditional PID matrix.

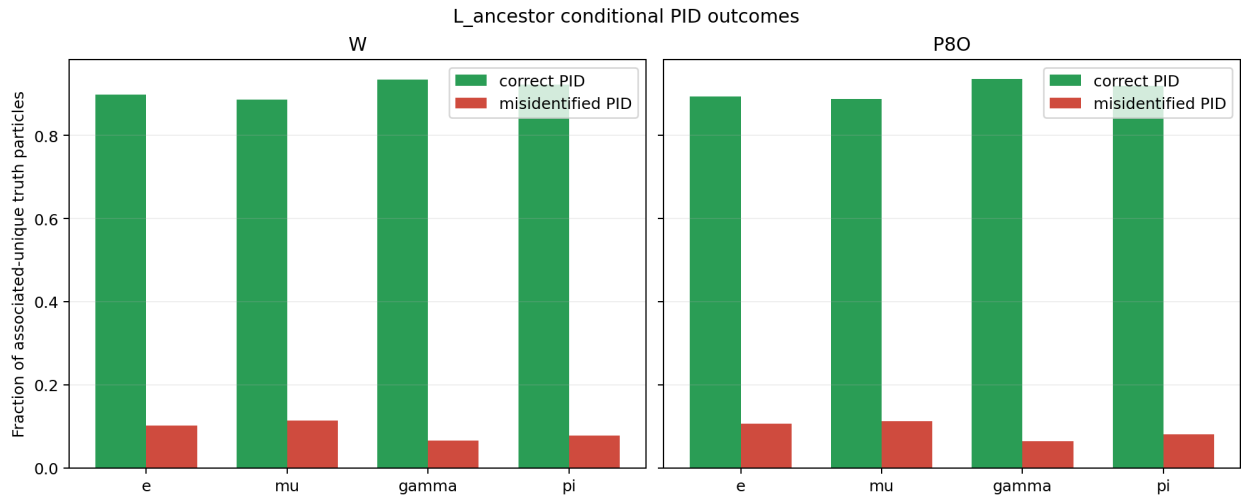
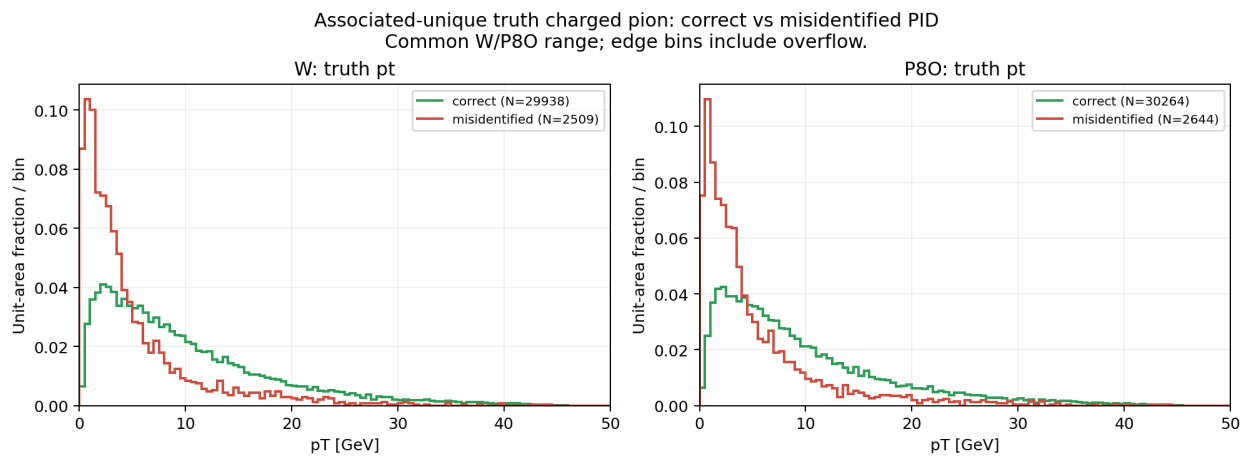
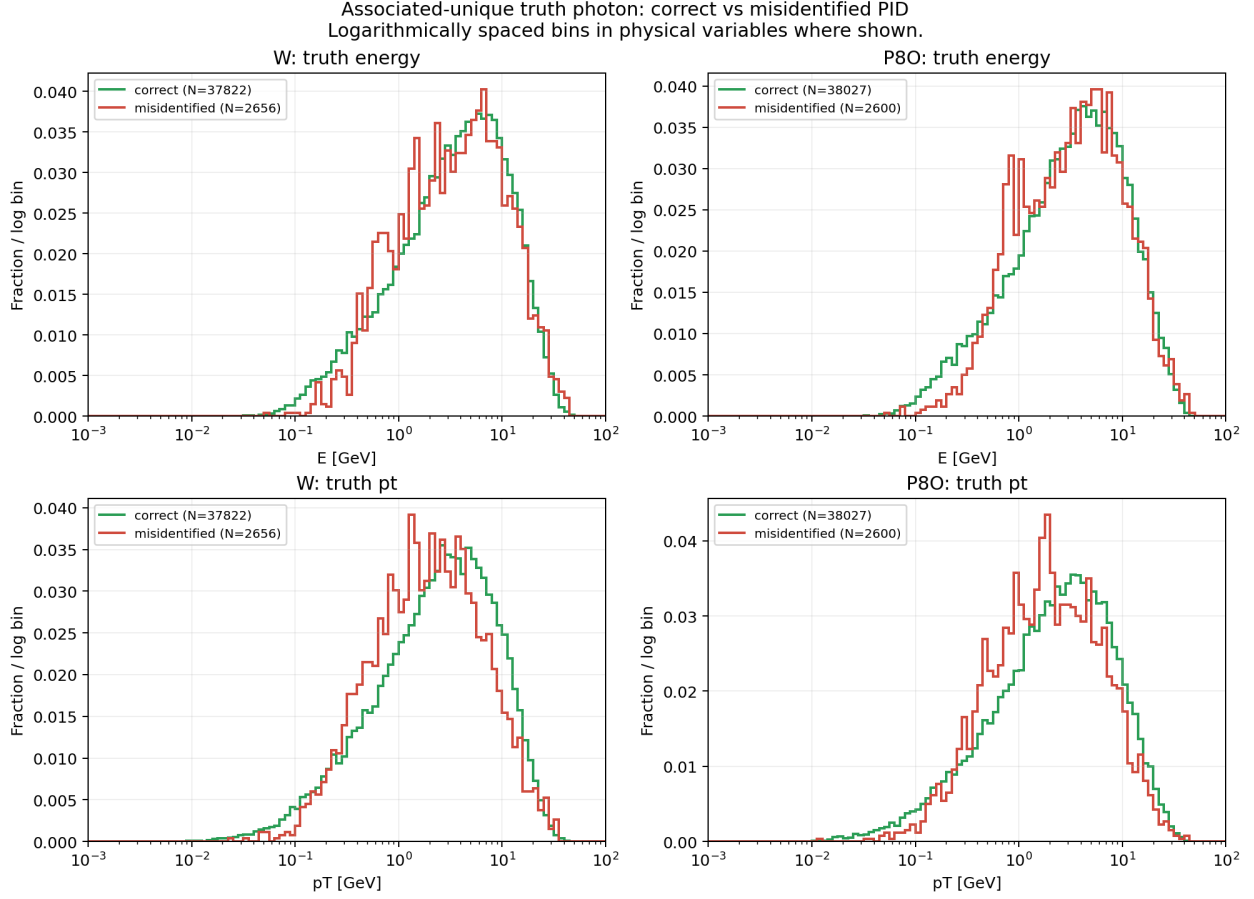
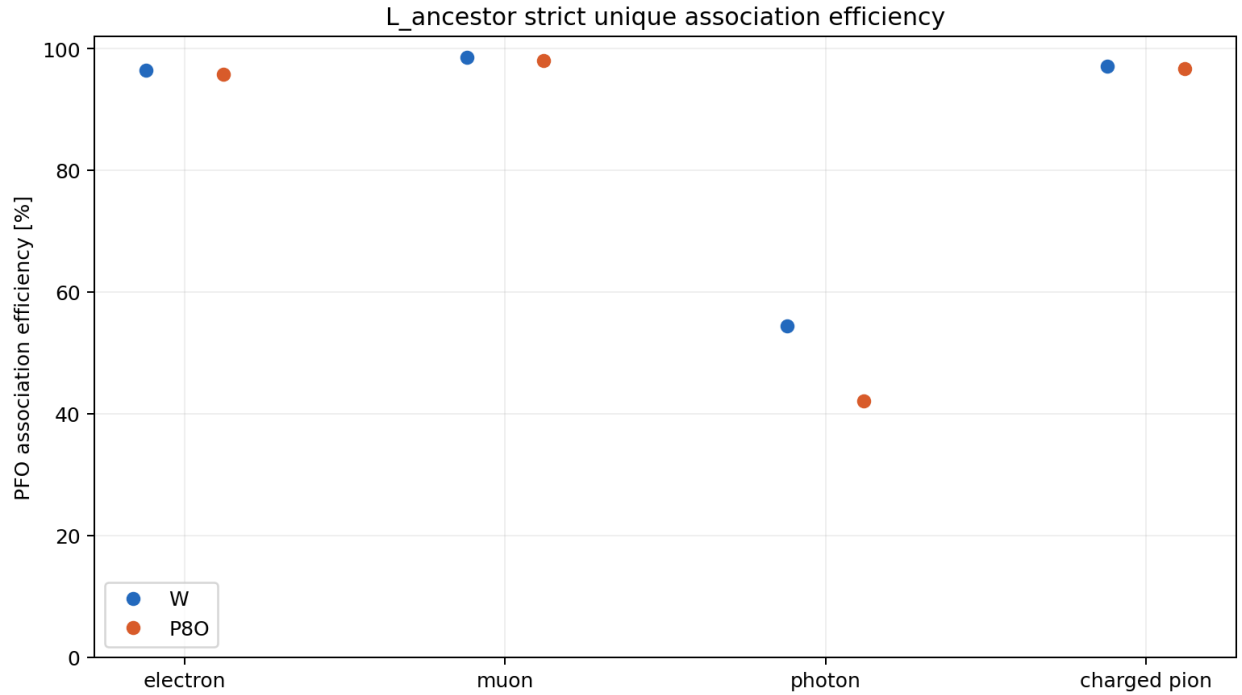


Figure 15: Conditional correct and misidentified fractions.

Figure 16: Selected charged-pion p_T : correct PID versus mis-ID.

Figure 17: Selected-photon energy and p_T : correct PID versus mis-ID.Figure 18: L_{ancestor} PFO association efficiency by selected-truth species.

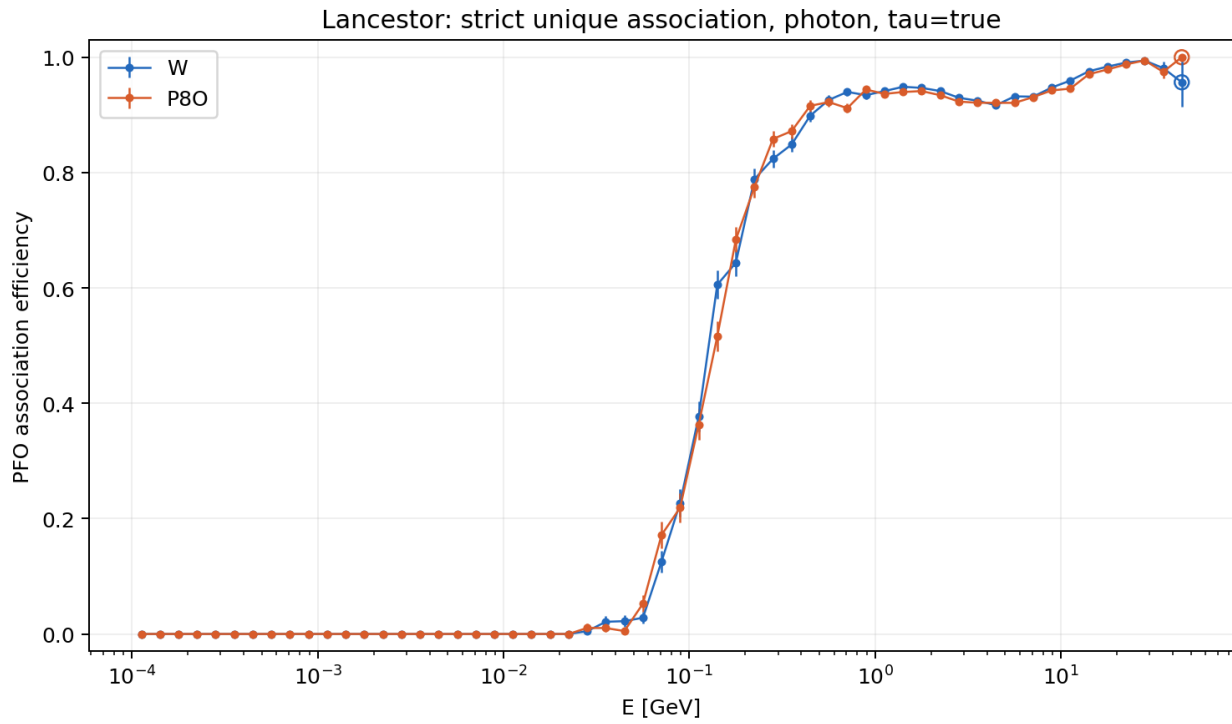


Figure 19: Tau-origin photon association efficiency versus physical energy.

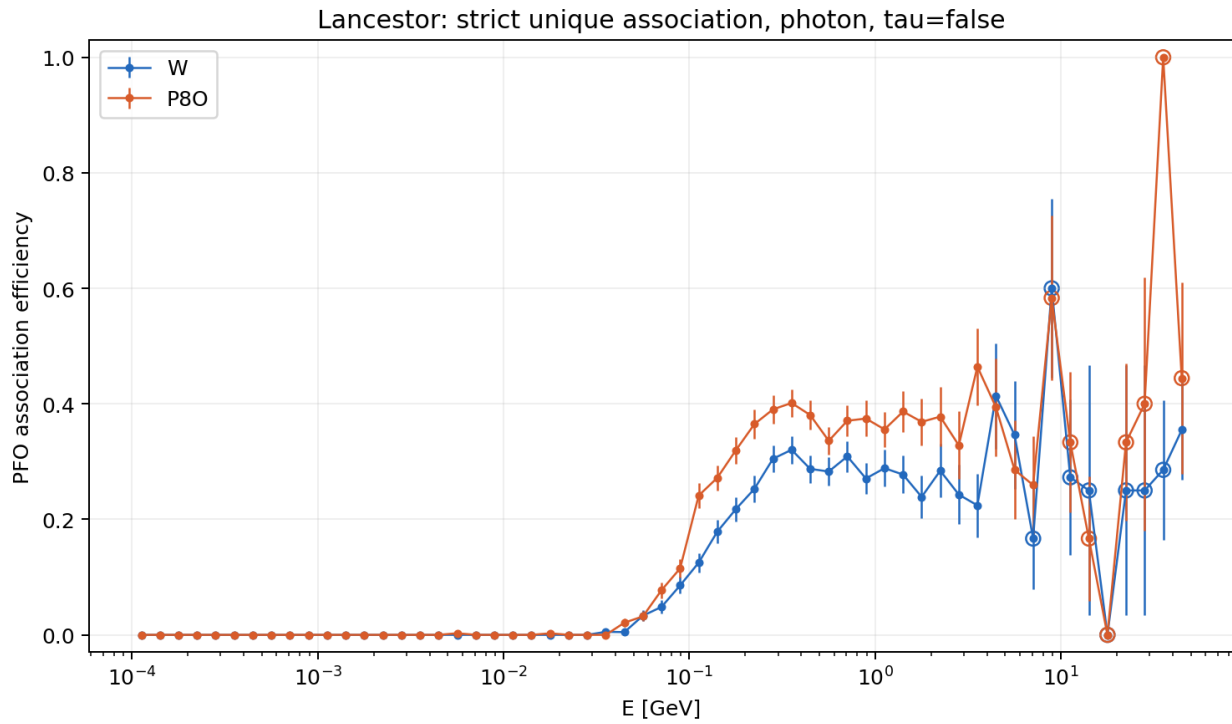


Figure 20: Non-tau photon association efficiency versus physical energy.

Charged pions are the control: association is about 97 %, conditional PID about 92 %, and W/P8O differences are small. Their main problematic region is low p_T and strongly forward/backward phase space.

9 Photon origin and event-record semantics

P8O has 49,120 selected photons with a same-sign electron-copy ancestry chain ending on parentless incoming beam e^-/e^+ objects with approximately $E = 45.595$ GeV and $|\cos\theta| = 0.9999755$. This is positive evidence for `explicit_ISR`.

All 25,879 W non-tau photons are parentless. Beam objects exist, but no photon-to-beam ancestry is stored; they remain `parentless_unresolved`. The two populations are partially similar in being soft and forward/backward, but P8O is substantially softer and more beam-collimated. Kinematics do not establish physical equivalence.

Table 8: Frozen photon-origin populations.

photon_origin_v1	W	P8O
parentless_unresolved	25,879	0
explicit_ISR	0	49,120
electron_parent_non_tau_unresolved	0	1
tau_decay_pi0	39,585	39,727
tau_direct_daughter_unresolved	8,640	7,697
tau_other_descendant	240	80

Direct tau daughters number 8,640 W and 7,697 P8O and usually appear beside continuing tau copies. The mixture of radiation/bookkeeping and terminal decay configurations does not provide a positive-evidence, cross-generator FSR definition. They remain `tau_direct_daughter_unresolved`.

10 Origin $\rightarrow$ association $\rightarrow$ conditional PID

Table 9: L_ancestor origin-factorized photon performance. Each cell is W/P8O.

Origin pair	Association	Conditional photon PID	Full photon outcome
parentless / explicit ISR	4.695/3.530 %	94.239/94.348 %	4.424/3.330 %
π^0 descendant	91.350/90.757 %	93.393/93.491 %	85.316/84.845 %
direct tau daughter	33.854/36.001 %	93.573/94.551 %	31.678/34.040 %

Once a unique PFO exists, photon PID is stable at roughly 93–95 % across samples and origins. The large variation occurs upstream, in truth population, phase space, and association.

For context, W parentless efficiencies under G/L_direct/L_ancestor are 4.475/3.938/4.695 % and P8O explicit-ISR efficiencies are 3.361/3.013/3.530 %. These are association-definition dependence, not detector-efficiency variation.

11 Frozen scientific interpretation

The large global W/P8O photon difference is primarily driven by the different non-tau photon populations and their phase-space composition. P8O contains an explicit-ISR population that is substantially softer and more beam-collinear than the unresolved W parentless population. Tau-decay photon

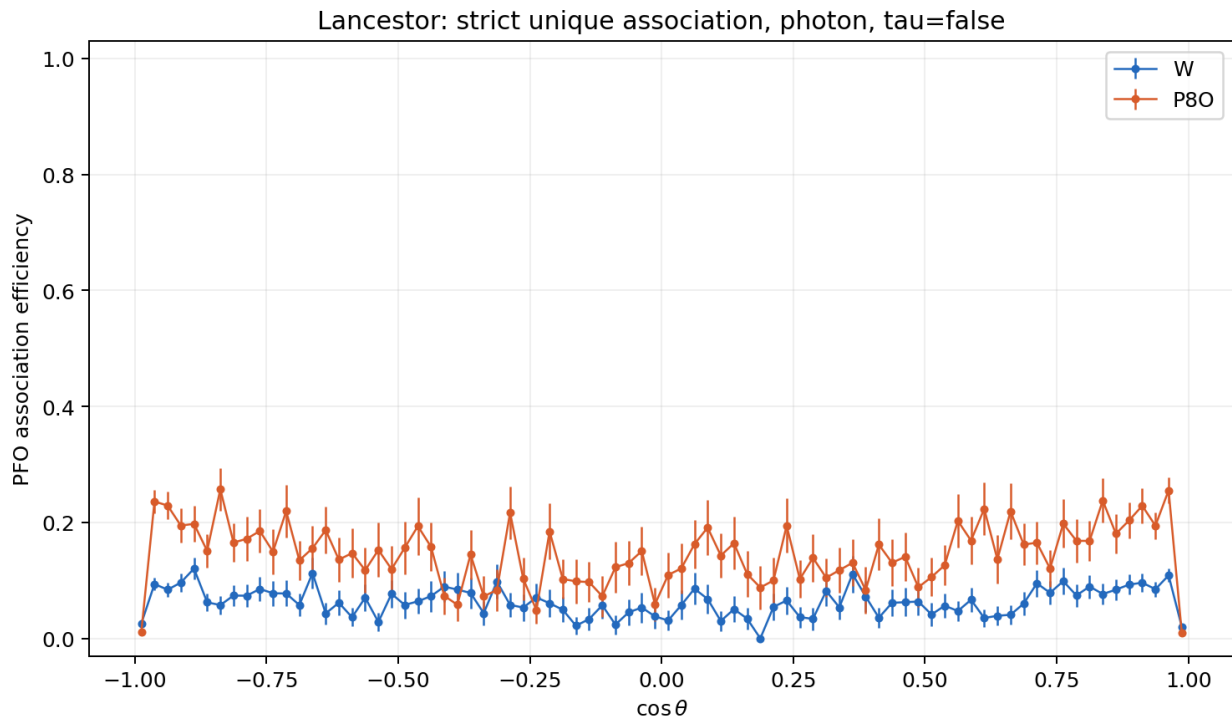


Figure 21: Non-tau photon association efficiency versus absolute polar cosine.

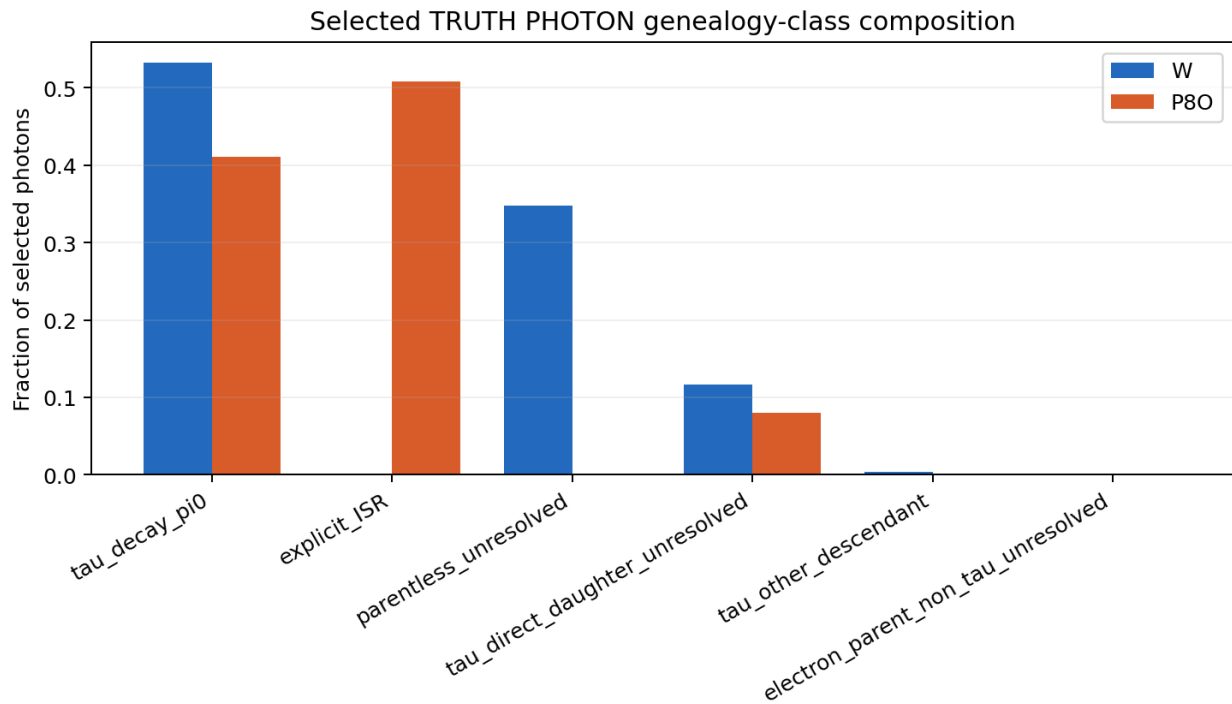


Figure 22: Frozen photon-origin composition.

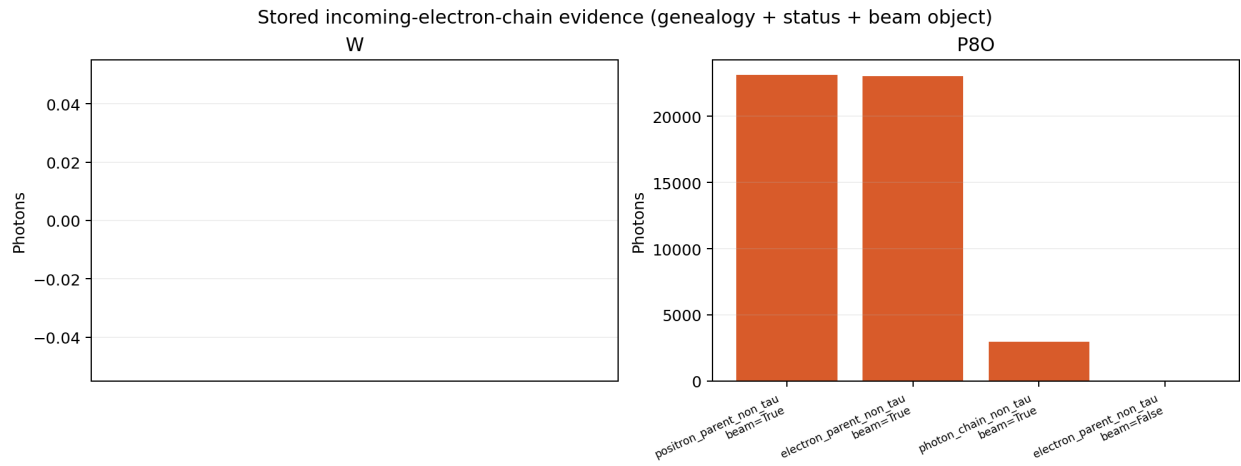


Figure 23: Stored incoming-electron-chain evidence. It supports P80 explicit ISR, not a W ISR label.

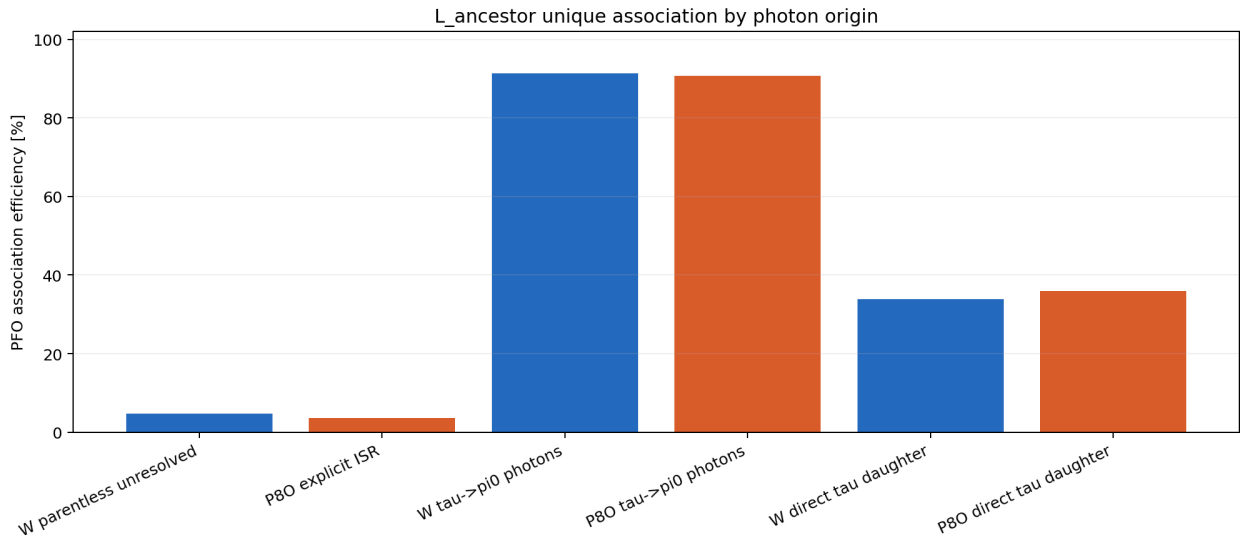


Figure 24: L_ancestor association efficiency by frozen photon origin.

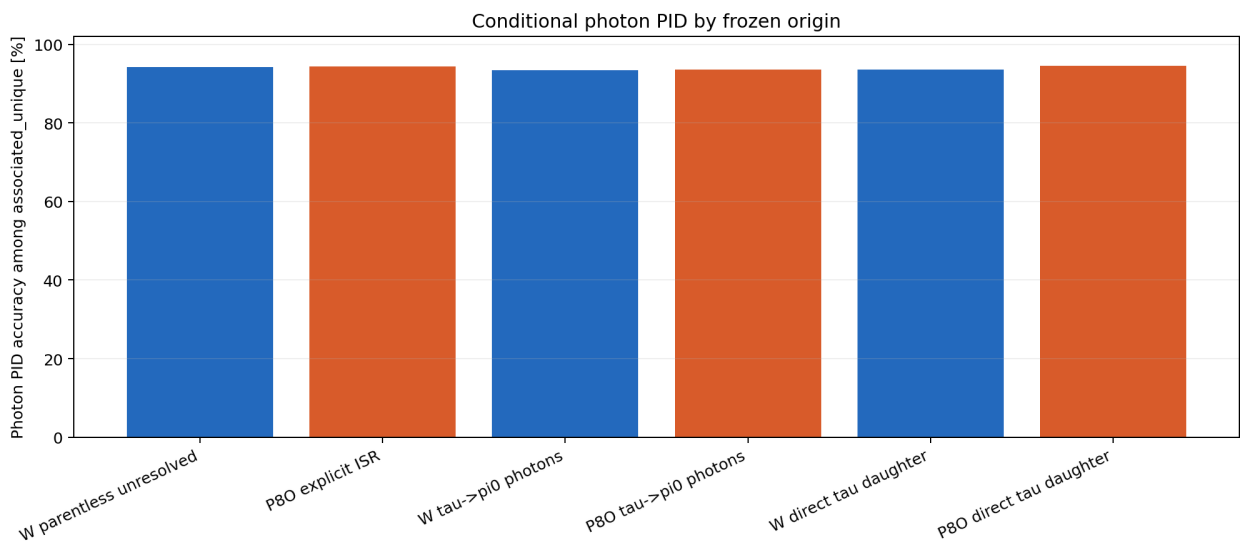


Figure 25: Conditional photon PID accuracy by origin.

populations have very similar truth kinematics and PFO association efficiencies. Once a unique PFO exists, conditional photon PID is very similar across samples and origins. The global migration difference is therefore dominated by **sample/generator truth composition plus truth-to-PFO association/reconstructability**, not a large difference in reconstructed photon PID.

12 Explicit limitations and freeze declaration

1. W `parentless_unresolved` photons cannot be called ISR.
2. P8O `explicit_ISR` is supported by stored generator genealogy.
3. tau `direct_daughter_unresolved` photons cannot currently be called FSR.
4. G/L `direct`/L `ancestor` are association semantics, not detector definitions.
5. This is not a generator-physics validation of WHIZARD versus PYTHIA8.
6. Raw all-photon outcomes cannot establish that one reconstruction globally reconstructs photons better.
7. Equal-statistics detailed audits use only 18,000 events per sample.

These are limitations of stored event-record information, not pending software bugs.

STATUS: SCIENTIFIC ANALYSIS FROZEN AT 2026-09-01
Frozen scope: W/P8C/P8O migration matrices; equal-statistics W/P8O tau topology; PID; PFO association efficiency; photon-origin discovery; and photon-origin reconstruction/PID factorization.

The individual reports, machine-readable tables, validation summaries, relevant scripts, and selected figures remain authoritative. Their sizes and SHA-256 hashes are recorded in `artifacts/fcc_tau_w_p8o_consolidated_freeze_20260901/`. No simulation, reconstruction, generator, linker, HitAnalysis, or scientific audit was rerun. No scientific definition changed, and no derived output was written to Lustre.

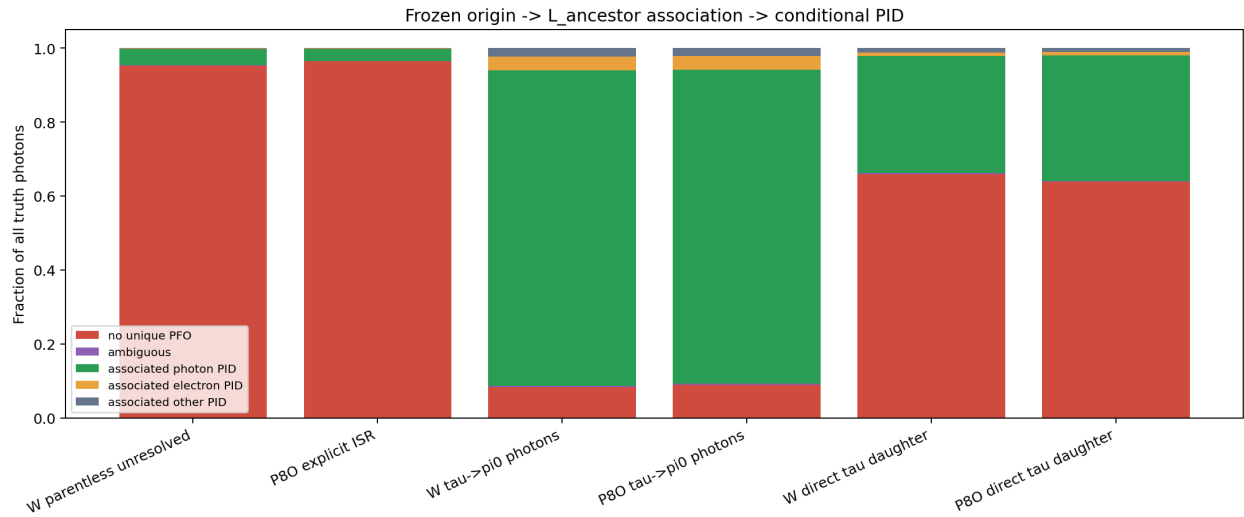


Figure 26: Full truth-origin $\rightarrow$ association $\rightarrow$ reconstructed-PID outcome.